

Assignment

Problem 1:

A single-slope ADC integrates a reference voltage (using ramp generator) until it matches the input analog voltage. Suppose the clock frequency is 50 kHz, the counter is 8-bit, and the reference ramp has a slope of 1.953 V/ms.

- (i) Find the quantization step size of the ADC in volts. (The quantization step size is the smallest input change that results in one count)
- (ii) Calculate the digital output for an input voltage of 3.91 V.
- (iii) Determine the maximum conversion time (The time an ADC takes for conversion when the input is full-scale, i.e., when the counter reaches its maximum count)

Problem 2:

A dual-slope ADC operates in two phases:

Integration (Run-Up): The input voltage is applied to an integrator for a fixed time. During this period, the counter runs at the clock frequency.

De-Integration (Run-Down): A reference voltage of opposite polarity is applied to the integrator, causing the output to ramp back down to zero. The counter continues until the output reaches zero. The time (and hence counts) taken during the run-down is proportional to the input voltage.

Consider a dual-slope ADC, where the input voltage is integrated (run-up) for a fixed time of 40.96 ms. The clock frequency is 100 kHz, and the reference voltage is 5 V. For an input of 2.41 V:

- (i) Calculate the number of counts accumulated during the integration phase (run-up).
- (ii) Determine the number of bits in the counter.
- (iii) Find the de-integration time required for the reference to bring the ramp output back to zero.
- (iv) Determine the digital output code (decimal and binary) corresponding to the 2.41 V input.

Problem 3:

A system designer compares a single-slope and a dual-slope ADC, both uses a 12-bit counter and a clock frequency of 1 MHz . The analog input is 4.5 V and the reference is 5 V for both cases.

- (i) Calculate the conversion time for in the given input voltage for each architecture.
- (ii) Find the digital output code in both cases.
- (iii) Comment on which ADC is more suitable for high-accuracy measurement in noisy environments.